NAME: _____ Geometry (FL B.E.S.T.)

## 8.2 Weighted Average of Points (Section Formula)

A point $P$ that divides $\overline{AB}$ in the ratio $AP : PB = m : n$ is a **weighted average** of the endpoints — it sits closer to whichever endpoint has the larger share of the ratio. Since $P$ is a fraction $\frac{m}{m+n}$ of the way from $A$ to $B$, you can find it with $P = A + \frac{m}{m+n}(B - A)$, which expands to the section formula $P = (\frac{nx_A + mx_B}{m+n}, \frac{ny_A + my_B}{m+n})$. When $m = n$ (ratio $1 : 1$), this collapses to the ordinary midpoint formula — the midpoint is just the special case of a $1 : 1$ weighted average. This worksheet also runs the formula in reverse: given the partition point and one endpoint, find the missing endpoint.

## Practice Set — All Questions

15 questions. Work them on your own paper, then check the key below.

**1.** $A(2, 4)$, $B(8, 10)$. Midpoint?

**2.** $A(0, 0)$, $B(9, 6)$; $AP : PB = 1 : 2$. Find $P$.

**3.** $A(1, 1)$, $B(4, 7)$; $AP : PB = 2 : 1$. Find $P$.

**4.** $\frac{1}{4}$ of the way from $(0, 0)$ to $(12, 8)$.

**5.** $\frac{3}{4}$ of the way from $(0, 0)$ to $(12, 8)$.

**6.** $A(0, 0)$, $B(10, 15)$; $AP : PB = 3 : 2$. Find $P$.

**7.** What fraction of the way is ratio $1 : 3$ from $A$?

**8.** $A(-4, 2)$, $B(8, 8)$; $AP : PB = 1 : 2$. Find $P$.

**9.** $A(2, 0)$, $B(6, 8)$; $AP : PB = 3 : 1$. Find $P$.

**10.** Ratio $1 : 1$ gives the...

**11.** $A(5, 5)$, $B(5, 17)$; halfway point?

**12.** $\frac{2}{5}$ of the way from $(1, 2)$ to $(11, 12)$.

**13.** $A(0, 6)$, $B(10, 1)$; $AP : PB = 2 : 3$. Find $P$.

**14.** Point $P(4, 5)$ divides $\overline{AB}$ so that $AP : PB = 3 : 5$, with $A(1, 2)$. Find $B$.

**15.** $A(2, -1)$, $B(14, 11)$; $AP : PB = 3 : 1$. Find $P$, then verify $AP + PB = AB$ using the distance formula.

## ✓ Answer Key

1. $(5, 7)$
2. $(3, 2)$
3. $(3, 5)$
4. $(3, 2)$
5. $(9, 6)$
6. $(6, 9)$
7. $\frac{1}{4}$
8. $(0, 4)$
9. $(5, 6)$
10. Midpoint
11. $(5, 11)$
12. $(5, 6)$
13. $(4, 4)$
14. $B = (9, 10)$
15. $P = (11, 8)$; $AP = 9\sqrt{2}$, $PB = 3\sqrt{2}$, $AB = 12\sqrt{2}$, and $9\sqrt{2} + 3\sqrt{2} = 12\sqrt{2}$